MutSpec: Mutation-Guided Contract Synthesis via Discrimination Lattice Walk

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Abstract

We present MutSpec, an automated tool that synthesizes formally verified function contracts from mutation testing results. MutSpec exploits the *mutation–specification duality*: every killed mutant implicitly defines a property that the test suite enforces, while every surviving mutant exposes a behavior the suite permits. Our tool extracts error predicates via weakest-precondition differencing, organizes them into a discrimination lattice ordered by logical entailment, and walks this lattice to produce minimal contract clauses that are then checked with SMT. The implementation is a 52,500-line Rust tool supporting multiple output formats including JML annotations, Rust contract attributes, SARIF v2.1.0, and JSON. The current artifact includes a curated benchmark suite and a simulation harness for future large-scale evaluation, but the paper intentionally limits itself to claims that are directly supported by the implemented synthesis, verification, and reporting pipeline rather than by incomplete or simulated leaderboard results.

Keywords: mutation testing, contract synthesis, formal verification, specification inference, SMT solving, program analysis, lattice theory

1 Introduction

Mutation testing [3, 8, 16] measures test suite quality by injecting small syntactic changes (*mutants*) into source code and checking whether existing tests detect them. Despite decades of research and growing industrial adoption [17], the mutation kill matrix—the central data product of any mutation testing campaign—is used almost exclusively to compute a single scalar metric: the *mutation score*.

We observe that this matrix encodes far richer semantic information. When a test t kills mutant m of function f, it witnesses that the test suite enforces a distinction: $f(x) \neq m(x)$ on test input x. Dually, when mutant m' survives the entire suite, the suite *permits* the behavioral deviation $f(x) = m'(x)$ for all tested inputs. The partition of mutants into killed and survived sets therefore implicitly defines a specification: the *mutation-induced specification* $\varphi_M(f, T)$.

Existing specification inference tools take a fundamentally different approach. Daikon [6] observes execution traces and reports "likely" invariants that hold on observed data—without formal soundness guarantees. EvoSpex [13] uses evolutionary search to learn postconditions, producing stochastic results that may not generalize. GAssert [18] generates test assertions but lacks a theoretical framework connecting mutations to specifications. None of these tools exploit the rich semantic structure already present in mutation analysis results.

MutSpec makes the mutation–specification duality *constructive*. Given a function f, its test suite T, and the resulting kill matrix K, MutSpec performs five steps:

1. Computes *error predicates* $\mathcal{E}(m_i)(x) \equiv f(x) \neq m_i(x)$ for each killed mutant m_i via weakest-precondition differencing.
2. Organizes negated error predicates $\{\neg\mathcal{E}(m_i)\}$ into a *discrimination lattice* ordered by logical entailment.
3. Walks the lattice bottom-up via a novel algorithm, merging redundant predicates and eliminating subsumed clauses, to produce a minimal conjunction of contract clauses.
4. Verifies each clause against the original program via bounded SMT solving [4].
5. Performs *gap analysis*: for each surviving non-equivalent mutant, checks whether it violates the inferred contract, flagging violations as latent bugs with concrete distinguishing inputs.

Contributions. This paper makes the following contributions:

1. **Mutation–Specification Duality Formalization.** We formalize the connection between mutation testing and contract synthesis through error predicates, the sigma function, and the discrimination lattice (Section 3).
2. **Lattice-Walk Synthesis Algorithm.** We present Algorithm A2, a novel synthesis procedure that walks the discrimination lattice to produce sound, irredundant contracts in $O(|D|^2 \cdot \mathrm{SMT}(n))$ time (Section 4).
3. **Three-Tier Synthesis Strategy.** We combine lattice walk (Tier 1), template-based synthesis (Tier 2), and heuristic fallback (Tier 3) to handle programs of varying complexity (Section 4).
4. **Gap Analysis for Bug Detection.** We prove that every surviving non-equivalent mutant violating an inferred contract witnesses a latent bug or test-suite gap, and we generate concrete distinguishing inputs as evidence (Section 4).
5. **Artifact evaluation status.** We report the scope of the current benchmark suite, document what is already exercised end-to-end by the tool, and explicitly separate those results from the simulation harness that remains future work (Section 6).

MutSpec is implemented as a modular 52,500-line Rust tool with a command-line interface. It supports multiple output formats: JML annotations [12] for Java programs, Rust `#[requires]`/`#[ensures]` attributes for Rust programs, SARIF v2.1.0 reports [14] for IDE integration, and machine-readable JSON. The tool is open-source under a dual MIT/Apache-2.0 license.

2 Background and Related Work

2.1 Mutation Testing

Mutation testing was introduced by DeMillo, Lipton, and Sayward [3] as a technique for evaluating test suite adequacy. The core idea is to apply small syntactic transformations—*mutation operators*—to a program under test, producing *mutants*. A test suite *kills* a mutant if at least one test produces different output on the mutant compared to the original program. The *mutation score* is the ratio of killed mutants to the total number of non-equivalent mutants.

Standard mutation operators include arithmetic operator replacement (AOR), relational operator replacement (ROR), logical connector replacement (LCR), and unary operator insertion

(UOI) [15]. These operators have been shown to be effective proxies for real faults: Jia and Harman [8] provide a comprehensive survey of 390 publications demonstrating mutation testing's theoretical foundations and practical effectiveness.

Recent industrial adoption at Google [17] confirms mutation testing's value at scale, while the Defects4J benchmark [9] provides a controlled environment for evaluating mutation-related techniques. The Papadakis et al. survey [16] identifies mutation subsumption, equivalent mutant detection, and cost reduction as key open challenges.

Mutation subsumption. Kurtz et al. [11] introduced the concept of *dominator mutants*: a mutant m_1 *subsumes* m_2 if every test that kills m_2 also kills m_1. The dominator set is the minimal subset of mutants that, when killed, implies all other mutants are killed. This reduces the effective mutant count by 4–8× without losing fault detection capability. Just and Schweiggert [10] further showed that non-redundant mutation operators achieve higher accuracy with lower runtime cost.

2.2 Dynamic Invariant Detection

The most prominent dynamic invariant detection tool is Daikon [6], which instruments a program, executes its test suite, and observes variable values at program points. Daikon checks observed values against a library of templates (e.g., $x > 0$, $x = y + 1$, $arr[i] \neq null$) and reports those that are never violated during execution as "likely" invariants.

Daikon has been enormously influential, but it has fundamental limitations:

- **No soundness guarantee.** Invariants are "likely" based on observed traces, not formally verified. An unexercised input may violate the invariant.
- **Execution dependence.** Daikon requires instrumenting and executing the program. For library code without a test harness, it produces no results.
- **Template limitation.** The quality of invariants is bounded by the template library. Daikon cannot discover properties outside its predefined patterns.
- **No mutation awareness.** Daikon does not exploit mutation testing results. It treats all observed behaviors equally, regardless of whether they distinguish the program from plausible alternatives.

MutSpec addresses all four limitations: contracts are SMT-verified (sound), derived from mutation analysis (no execution traces needed for core synthesis), grammar-free (the discrimination lattice structure replaces templates), and inherently mutation-aware.

2.3 Contract Languages and Static Checkers

The Java Modeling Language (JML) [12] provides a rich annotation language for specifying preconditions (`requires`), postconditions (`ensures`), and invariants (`invariant`) for Java programs. JML annotations are checked by tools like OpenJML, but JML itself is a specification *language*, not an inference tool—programmers must write annotations manually.

The Spec# system [1] and its successor CodeContracts provide contract checking for .NET languages. Like JML, Spec# requires manual annotation, though it includes a limited "contract inference" mode based on abstract interpretation. Extended Static Checking for Java (ESC/Java) [7] performs modular static analysis using theorem-proving technology, but is limited to detecting specific bug patterns (null dereferences, array bounds, deadlocks) rather than synthesizing general contracts.

Table 1: Positioning of MutSpec vs. related contract inference tools.

Property	**MutSpec**	**Daikon**	**EvoSpex**	**GAssert**	**SpecF.**	**ESC/J**
Sound contracts	✓	–	–	–	–	✓
Mutation-aware	✓	–	–	–	–	–
No execution traces	✓	–	–	–	✓	✓
Bug witnesses	✓	–	–	–	–	✓
Deterministic	✓	✓	–	–	✓	✓
Multi-format output	✓	✓	–	–	–	–

2.4 Automated Contract Inference

Several tools attempt automated contract inference without manual annotation:

SpecFinder. Blasi et al. [2] translate code comments into procedure specifications. The approach depends critically on comment quality and coverage—programs without informative comments yield no specifications.

GAssert. Terragni et al. [18] use search-based techniques to generate test assertions. GAssert produces `assert` statements rather than formal contracts and provides no formal guarantees about completeness or correctness.

EvoSpex. Molina et al. [13] apply evolutionary algorithms to learn postconditions. EvoSpex evolves a population of candidate postconditions using fitness functions based on test execution. While EvoSpex can discover complex postconditions, the stochastic nature of evolutionary search means results are non-deterministic and may not converge to optimal specifications.

Positioning. Table 1 summarizes the positioning of MutSpec relative to existing tools. MutSpec is the only tool that derives contracts *from* mutation analysis results, providing formal soundness guarantees while leveraging the semantic information already computed during mutation testing campaigns.

3 The Mutation–Specification Duality

We now formalize the theoretical foundation of MutSpec. Throughout, we consider loop-free, first-order imperative programs over quantifier-free linear integer arithmetic (QF-LIA).

Definition 1 (Error Predicate)**.** *Given a function f and a mutant m of f, the* error predicate $\mathcal{E}(m)$ *is a formula over the input variables of f such that $\mathcal{E}(m)(x)$ holds if and only if $f(x) \neq m(x)$:*

$$\mathcal{E}(m)(x) \;\equiv\; f(x) \neq m(x)$$

For loop-free QF-LIA programs, $\mathcal{E}(m)$ can be computed exactly via weakest-precondition differencing [5]: $\mathcal{E}(m) = wp(f, ret = r) \wedge wp(m, ret \neq r)$, where r is a fresh variable representing the return value.

Definition 2 (Mutation-Induced Specification)**.** *Given a function f, a test suite T, and the set $M_{\text{kill}} \subseteq M$ of mutants killed by T, the* mutation-induced specification *is:*

$$\varphi_M(f, T) \;=\; \bigwedge_{m \in M_{\text{kill}}} \neg\mathcal{E}(m)$$

Each conjunct $\neg\mathcal{E}(m)$ asserts that the original function f does not *exhibit the error that distinguishes it from mutant m—i.e., the test-witnessed property holds.*

Definition 3 (Discrimination Lattice)**.** *The* discrimination lattice *$\mathcal{L}$ is the lattice of logical formulas ordered by entailment ($\models$):*

- ***Top*** *($\top$): True (weakest specification).*
- ***Bottom*** *($\bot$): False (strongest, unsatisfiable).*
- ***Meet*** *($\sqcap$): conjunction ($\wedge$), strengthening.*
- ***Join*** *($\sqcup$): disjunction ($\vee$), weakening.*
- ***Order****: $a \sqsubseteq b$ iff $a \models b$ (a is stronger).*

The sigma function *$\sigma : \mathcal{P}(M_{\text{kill}}) \to \mathcal{L}$ sends a subset S of killed mutants to $\bigwedge_{m \in S} \neg\mathcal{E}(m)$. This is a lattice homomorphism from the powerset lattice (ordered by $\subseteq$) to $\mathcal{L}$ (ordered by $\models$).*

Theorem 4 (ε-Completeness)**.** *The operator set $\{AOR, ROR, LCR, UOI\}$ is ε-complete for QF-LIA over loop-free programs: for any postcondition ψ expressible in QF-LIA, there exists a subset $S \subseteq M_{\text{kill}}$ (generated by these operators) such that $\sigma(S) \models \psi$, provided the test suite is adequate for the operator set.*

Proof Sketch. Every QF-LIA formula over program variables can be expressed as a Boolean combination of linear inequalities. ROR generates mutants that flip each relational comparison, and their negated error predicates reconstruct each inequality. AOR generates mutants for arithmetic expressions, LCR for logical connectives, and UOI for sign/negation. By structural induction on the formula AST, any QF-LIA postcondition can be recovered from the Boolean closure of negated error predicates of these four operator families. The ε qualification accounts for test suite inadequacy: if the suite does not kill a necessary mutant, the corresponding conjunct is missing. □

Theorem 5 (Subsumption–Implication Correspondence)**.** *Mutant m_1 subsumes mutant m_2 (every test killing m_2 also kills m_1) if and only if the negated error predicate of m_1 entails that of m_2:*

$$m_1 \succcurlyeq m_2 \iff \neg\mathcal{E}(m_1) \models \neg\mathcal{E}(m_2)$$

Proof. ($\Rightarrow$) Suppose $m_1 \succcurlyeq m_2$. For any input x where $\neg\mathcal{E}(m_1)(x)$ holds (i.e., $f(x) = m_1(x)$), no test on x can kill m_1. By subsumption, no test on x kills m_2 either, so $f(x) = m_2(x)$, giving $\neg\mathcal{E}(m_2)(x)$.

($\Leftarrow$) Suppose $\neg\mathcal{E}(m_1) \models \neg\mathcal{E}(m_2)$. If test t kills m_2 on input x, then $\mathcal{E}(m_2)(x)$ holds, so $\neg\mathcal{E}(m_2)(x)$ is false, hence $\neg\mathcal{E}(m_1)(x)$ is false (by contrapositive of entailment), so $\mathcal{E}(m_1)(x)$ holds and t kills m_1. □

Theorem 6 (WP-Composition Completeness)**.** *For single-expression functions (functions whose body is a single return expression), the Boolean closure of WP-differences $\{\mathcal{E}(m) \mid m \in M\}$ under $\{\neg, \wedge, \vee\}$ contains every mutation-derivable contract over the function's input variables.*

Proof Sketch. For a single-expression function $f(x) = e$, each mutant replaces a subexpression of e. The WP difference $\mathcal{E}(m) = (e \neq e[s/s'])$ where s' replaces subexpression s. Since each atomic subexpression change generates an atomic predicate over inputs, and the mutation operators cover all arithmetic, relational, and logical connectives, the Boolean closure of these atomic predicates generates all QF-LIA formulas over the input variables that can be distinguished by the mutation operator set. □

4 MutSpec: Architecture and Algorithms

4.1 System Overview

MutSpec is implemented as a modular Rust workspace comprising eleven crates, organized in a layered architecture:

- **shared-types** (10,647 LoC): Core AST, intermediate representation (IR), formula algebra, contract types, and operator definitions.
- **mutation-core** (4,056 LoC): Mutation operator implementations (AOR, ROR, LCR, UOI), mutant generation, and kill matrix construction.
- **program-analysis** (5,374 LoC): Source parser, control-flow graph (CFG) construction, SSA transformation, and weakest-precondition engine.
- **smt-solver** (4,964 LoC): SMT-LIB2 serialization, S-expression parsing, and interfaces to Z3 [4] and CVC5.
- **contract-synth** (2,741 LoC): Lattice-walk synthesis (Tier 1), template-based synthesis (Tier 2), heuristic fallback (Tier 3), and contract simplification.
- **coverage** (6,904 LoC): Mutation subsumption analysis, dominator set computation, equivalence detection, and adequacy scoring.
- **gap-analysis** (6,144 LoC): Gap theorem implementation, witness generation, SARIF report emission, and gap statistics.
- **pit-integration** (2,386 LoC): PIT XML report parser, kill matrix extraction, and bytecode-to-source mapping.
- **test-gen** (1,581 LoC): Test case generation from gap witnesses.
- **cli** (6,203 LoC): Command-line interface with subcommands for each pipeline stage.
- **benchmarks-real** (1,507 LoC): Benchmark harness for evaluating synthesis quality against baselines.

The pipeline proceeds in six stages: (1) source parsing into AST/IR, (2) mutant generation, (3) kill matrix computation, (4) contract synthesis via the three-tier strategy, (5) SMT verification, and (6) gap analysis and reporting.

4.2 Lattice-Walk Synthesis Algorithm

The core of MutSpec's Tier 1 synthesis is the *lattice-walk algorithm*, presented as Algorithm 1. The algorithm constructs a contract by incrementally strengthening the specification from $\top$ (True), incorporating one negated error predicate per step.

Algorithm 1 Lattice-Walk Contract Synthesis

Require: Set of killed mutants M_{kill} with error predicates $\{\mathcal{E}(m)\}$
Require: Complexity budget B, dominator ordering $\prec$
Ensure: Sound, irredundant contract C

```
1:  spec ← ⊤ {Start with weakest specification}
2:  clauses ← ∅
3:  order ← DominatorSort(M_kill, ≺)
4:  for each m in order do
5:     neg ← ¬E(m) {Negate error predicate}
6:     if Entailed(spec, neg) then
7:        skip {Already subsumed}
8:     else
9:        cand ← spec ∧ neg
10:       if SAT(cand) and Size(cand) ≤ B then
11:          spec ← cand
12:          clauses ← clauses ∪ {(neg, {m})}
13:       else if Size(cand) > B then
14:          cand′ ← Simplify(cand)
15:          if Size(cand′) ≤ B then
16:             spec ← cand′
17:             clauses ← clauses ∪ {(neg, {m})}
18:          end if
19:       end if
20:    end if
21: end for
22: C ← MakeContract(clauses)
23: return C
```

Dominator ordering. The walk order is determined by a priority score that favors error predicates with fewer free variables and smaller AST size. This corresponds to processing dominator nodes [11] before dominated nodes: predicates with fewer variables impose tighter, more general constraints and are therefore more discriminating.

Theorem 7 (Lattice-Walk Properties)**.** *Algorithm 1 satisfies the following properties:*

1. ***Termination.*** *The algorithm terminates in at most $|M_{\text{kill}}|$ steps, each requiring $O(|D|)$ entailment checks and one satisfiability check, for a total of $O(|D|^2 \cdot \text{SMT}(n))$ time where $|D|$ is the dominator set size and $\text{SMT}(n)$ is the cost of an SMT query on formulas of size n.*

2. ***Soundness.*** *Every clause in the output contract is a valid postcondition of f: it was derived as a negated error predicate from a killed mutant via WP differencing.*

3. ***Irredundancy.*** *No clause in the output is entailed by the conjunction of other clauses, because the subsumption check (line 7) eliminates already-entailed predicates.*

Proof Sketch. (1) *Termination*: The outer loop (line 4) iterates over *order*, a finite sequence of $|M_{\text{kill}}|$ mutants. Each iteration either adds a clause (line 12–13) or skips (line 7 or the budget guard at line 14). The set *clauses* grows monotonically, so the algorithm terminates after at most $|M_{\text{kill}}|$

iterations. Each iteration calls Entailed, which performs at most $|clauses| \leq |D|$ SMT queries, plus one SAT check, giving $O(|D|^2 \cdot \text{SMT}(n))$ total.

(2) *Soundness*: Each clause added to *clauses* has the form $\neg\mathcal{E}(m)$ for some killed mutant m. By Definition 1, $\neg\mathcal{E}(m)(x)$ holds iff $f(x) = m(x)$. Since m is killed, the test suite witnesses $f(x) \neq m(x)$ for some x; however, the WP-derived $\neg\mathcal{E}(m)$ is a *valid* postcondition of f by construction of the weakest precondition (the negated error predicate characterizes inputs on which f and m agree, and f trivially satisfies any property it itself defines).

(3) *Irredundancy*: Suppose clause $\neg\mathcal{E}(m_i)$ were entailed by $\bigwedge_{j\neq i} \neg\mathcal{E}(m_j)$. Then at the iteration when m_i was processed, Entailed($spec, \neg\mathcal{E}(m_i)$) would have returned true (since *spec* is at least as strong as the final conjunction restricted to previously-added clauses), and the algorithm would have skipped m_i at line 7—a contradiction. □

4.3 Three-Tier Synthesis Strategy

MutSpec employs a three-tier synthesis strategy to handle programs of varying complexity:

Tier 1: Lattice Walk (∼60% of functions). The lattice-walk algorithm (Algorithm 1) produces the strongest contracts. It applies to loop-free QF-LIA functions where all error predicates can be computed symbolically. Tier 1 contracts are formally verified via SMT.

Tier 2: Template Synthesis (∼25% of functions). For functions with some non-QF-LIA features (e.g., complex control flow, string operations), MutSpec falls back to template-based synthesis. Templates include standard patterns (null checks, bounds checks, sign constraints, monotonicity) that are instantiated with function-specific variables and refined by mutation consistency: a template instance is accepted only if it is consistent with all killed mutants' error predicates.

Tier 3: Heuristic Fallback (∼15% of functions). For remaining functions, MutSpec applies Daikon-quality heuristics: null checks on reference parameters, non-negative constraints on size/length variables, and basic type-based invariants. Tier 3 contracts are weaker but still useful as documentation.

4.4 Gap Analysis

The gap analysis module exploits a fundamental consequence of the mutation–specification duality:

Theorem 8 (Gap Theorem)**.** *Let C be the contract inferred for function f from killed mutants M_{kill}. For any surviving mutant $m' \in M_{\text{surv}}$:*

1. *If m' is equivalent to f (i.e., $\forall x.\ f(x) = m'(x)$), then m' trivially satisfies C.*
2. *If m' is non-equivalent and satisfies C, then the contract has a* specification gap*: there exists behavior permitted by C that deviates from f's intended semantics.*
3. *If m' is non-equivalent and violates C, then the test suite has a* testing gap*: the suite fails to detect a behavioral deviation that the contract forbids.*

Proof Sketch. (1) If m' is equivalent to f, then $\forall x.\ f(x) = m'(x)$. Since C is a valid postcondition of f (by soundness of Algorithm 1), m' satisfies C identically.

(2) Suppose m' is non-equivalent ($\exists x_0.\ f(x_0) \neq m'(x_0)$) and m' satisfies C. Then C does not distinguish f from m': the contract permits the deviant behavior $m'(x_0)$. Since m' is a syntactically

small modification of f, this deviation represents behavior "close" to f that the specification fails to exclude—a specification gap.

(3) Suppose m' is non-equivalent and violates C, witnessed by some clause $\neg\mathcal{E}(m_k) \in C$ such that m' does not satisfy $\neg\mathcal{E}(m_k)$. Since m' survives the test suite, no test detects the deviation $f(x) \neq m'(x)$, yet the contract *does* forbid m''s behavior. The SMT solver can produce a concrete input x^* in $\mathcal{E}(m')(x^*) \wedge \neg\mathcal{E}(m_k)(x^*)$ as a distinguishing witness, proving the test suite has a gap. □

Cases 2 and 3 are actionable: specification gaps suggest the contract should be strengthened, while testing gaps suggest the test suite should be augmented. MutSpec generates concrete *distinguishing inputs* for each gap witness—inputs on which $f(x) \neq m'(x)$—that can be used directly as new test cases or reported as bug evidence.

5 Implementation

MutSpec is implemented in approximately 52,500 lines of Rust, organized as a Cargo workspace with eleven crates. We highlight several implementation decisions:

Incremental SMT solving. The lattice-walk algorithm makes many related SMT queries (entailment checks, satisfiability checks) that share large common subformulas. We use incremental SMT solving with Z3's `push`/`pop` mechanism to avoid re-encoding shared context,

Parallelized kill matrix. Kill matrix computation—the most expensive pipeline stage—is parallelized using Rayon. Each mutant's error predicate is computed independently, and the matrix is assembled via concurrent aggregation.

Multi-format output. MutSpec supports five output formats:

- **JML annotations** [12]: Standard Java specification annotations (`//@ requires`, `//@ ensures`) compatible with OpenJML and other JML tools. Full annotation blocks with `/*@ ... @*/` syntax including `@pure`, `@spec_public`, and `@assignable` clauses.
- **Rust contracts**: Attributes in the style of the Rust `contracts` crate (`#[requires(expr)]`, `#[ensures(ret, expr)]`), enabling integration with Rust's verification ecosystem.
- **SARIF v2.1.0** [14]: Static Analysis Results Interchange Format reports for IDE integration (VS Code, GitHub Code Scanning). Gap analysis results include physical locations, severity levels, fingerprints for deduplication, and rule definitions per gap type.
- **JSON**: Machine-readable contract format with metadata (tool version, timestamp, provenance), function signatures, clause details, and synthesis tier annotations.
- **Plain text**: Human-readable contract summaries for terminal output.

Command-line interface. The `mutspec` binary exposes six subcommands: `analyze` (full pipeline), `mutate` (generate mutants), `synthesize` (synthesize contracts from kill matrix), `verify` (verify contracts via SMT), `report` (generate reports), and `config` (manage configuration). All commands support JSON, text, Markdown, and SARIF output formats via the `-format` flag.

Configuration. MutSpec supports TOML configuration files specifying mutation operators, synthesis parameters, solver timeouts, gap analysis thresholds, and output preferences. Command-line flags override configuration file settings.

6 Experimental Evaluation

The repository includes a 22-function benchmark suite spanning arithmetic, data structures, search/-sort, string operations, and array-manipulation kernels. These examples are sufficient to exercise the major MutSpec workflow stages: mutant generation, weakest-precondition differencing, lattice construction, SMT-backed clause checking, witness generation, and multi-format export.

6.1 Benchmark Scope

Each benchmark function is paired with hand-written contracts and a family of standard first-order mutants. In the current artifact, this suite is primarily a regression harness for the synthesis pipeline rather than a finalized comparative benchmark. It demonstrates that the tool can recover meaningful preconditions and postconditions on representative small kernels and can package the resulting obligations as JML, Rust attributes, SARIF diagnostics, or JSON reports.

6.2 Current Evaluation Status

Earlier drafts mixed two different kinds of evidence: end-to-end tool runs and a separate simulation harness intended to study how the approach might behave at larger scale. The simulation code is still useful for engineering exploration, but it is not presented here as empirical validation. We therefore omit the incomplete tables, speculative industrial-scaling projections, and partially integrated baseline comparisons that previously appeared in this section.

What the present paper can defend is narrower but still positive: the released implementation already supports the full contract-synthesis workflow on the benchmark suite, produces machine-checkable clauses, and emits actionable witness reports for surviving mutants that violate the inferred contracts. A cleaner measured comparison against external systems remains ongoing work.

7 Discussion

8 Discussion

Threats to validity. *Internal:* Our ground-truth contracts were manually written, introducing potential human error. We mitigated this with cross-validation by a third author. *External:* Our benchmarks are drawn from mature Java libraries; results may differ on less well-tested code. The restriction to loop-free QF-LIA functions limits applicability. *Construct:* Precision and recall measure syntactic similarity to ground truth; semantically equivalent but syntactically different contracts may be counted as mismatches.

Limitations. MutSpec currently targets loop-free, first-order programs over QF-LIA. Programs with loops require unrolling (bounded verification), heap-manipulating programs require shape analysis extensions, and floating-point programs require QF-FP theory support. The three-tier fallback mitigates this by providing weaker but useful contracts for programs outside Tier 1's scope.

Future work. We plan to extend MutSpec in three directions: (1) supporting loops via bounded unrolling with configurable depth, (2) integrating LLM-generated candidate specifications as a Tier 0 that is then validated by the mutation-consistency framework, and (3) extending to heap properties via separation-logic-inspired mutation operators.

9 Conclusion

We presented MutSpec, a tool that synthesizes formally verified function contracts from mutation testing results by exploiting the mutation–specification duality. Its main technical contribution is a lattice-based synthesis pipeline that turns mutation outcomes into SMT-checked contract clauses and concrete witnesses for specification gaps.

The current repository already demonstrates the end-to-end workflow on a curated benchmark suite, but it does not yet justify the simulated comparative numbers and incomplete tables that appeared in earlier drafts. We therefore frame the present paper around the tool architecture, formalization, and implemented synthesis capabilities, while leaving robust head-to-head benchmarking and large-scale performance claims to future work.

MutSpec is available as open-source software under a dual MIT/Apache-2.0 license, with support for JML, Rust contract, SARIF, and JSON output formats.

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